Faculty of Engineering

Module Title: Engineering Chemistry

Sheet 3

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1. What is the sign of ΔH for an exothermic reaction? For an endothermic reaction?

2. Under what circumstances are ΔE and ΔH essentially equal?

3. Calculate the work done in joules by a chemical reaction if the volume increases from 3.2 L to 3.4 L against a constant external pressure of 3.6 atm. What is the sign of the energy change?

4. How much heat in kilojoules has evolved when 5.00 g of aluminum reacts with a stoichiometric amount of Fe_2O_3?

$$2\,Al(s) + Fe_2O_3(s) \longrightarrow 2\,Fe(s) + Al_2O_3(s) \qquad \Delta H^\circ = -852\ kJ$$

5. Calculate ΔH°_f in kJ/mol for benzene(C_6H_6), from the following data:

$$2\,C_6H_{6(l)} + 15\,O_{2(g)} \rightarrow 12\,CO_{2(g)} + 6\,H_2O_{(l)} \quad \Delta H^\circ = -6534\ kJ$$

$\Delta H^\circ_f\,(CO_2) = -393.5$ kJ/mol

$\Delta H^\circ_f\,(H_2O) = -285.8$ kJ/mol

6. Acetic acid (CH_3COOH), whose aqueous solutions are known as vinegar, is prepared by a reaction of ethyl alcohol (CH_3CH_2OH) with oxygen:

$$CH_3CH_2OH_{(l)} + O_{2(g)} \rightarrow CH_3CO_2H_{(l)} + H_2O_{(l)}$$

Use the following data to calculate the reaction: ΔH ° in kilojoules for the

$\Delta H^\circ_f[CH_3CH_2OH_{(l)}] = -277.7$ kJ/mol

$\Delta H^\circ_f[CH_3CO_2H(l)] = -484.5$ kJ/mol $\qquad$ $\Delta H^\circ_f[H_2O_{(l)}] = -285.8$ kJ/mol